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SPECTROSCOPY OF A BROWN DWARF CANDIDATE IN THE α PERSEI OPEN CLUSTER[1]

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ABSTRACT

We report intermediate-resolution spectroscopy in the range 625–775 nm of a faint ($R \sim 19$) very late type object, discovered in two CCD images taken near the center of the α Per cluster. The spectrum shows strong absorption molecular bands, Hα emission, and KI absorption typical of late-M dwarfs; Hα varies conspicously ($W_\lambda \sim 2.4$–0.8 nm) is a short time scale. The effective temperature and luminosity derived from the spectrum and *R,I* band photometry are consistent with those expected for an α Per (age ~50 Myr) 0.07 $M_\odot$ substellar object. We argue that in our object the lithium resonance line at 670.8 nm should be detectable with higher resolution, higher signal-to-noise ratio data, and we propose that this line is a potentially powerful spectroscopic discriminant between very low mass stars and brown dwarfs close to the substellar mass limit.

Subject headings: stars: individual: Ap 0323+4853 — stars: low-mass, brown dwarfs

1. INTRODUCTION

In the last few years, increasing observational efforts have been expended to detect isolated brown dwarfs (BDs), because of their natural interest in star formation and stellar evolution theory, and with the aim of assessing their potential contribution to the local mass density. Some free-floating BD candidates have been serendipitously discovered in the general star field, mainly because of their peculiar spectrum (strong molecular bands and emission of Hα), e.g., ESO 207−61 (Ruiz, Takamiya, & Roth 1991), the coolest dwarfs of Bessell (1991), and PC 0025+0447 (Schneider et al. 1991), but their masses are very uncertain due to a lack of knowledge about their age and distance. This is not a problem in the case of members of known galactic clusters and stellar associations. From theoretical models, we expect to detect BDs more easily while they are young (D'Antona & Mazzitelli 1985; Stringfellow 1991); hence, obvious places to search are nearby young clusters like the Pleiades and α Per. Direct CCD imaging surveys in the Pleiades (Jameson & Skillen 1989; Stauffer et al. 1989) have found a number of BD candidates. At least two have proper motions consistent with membership to the Pleiades, but the photometry is not entirely consistent with theoretical evolutionary tracks and isochrones (Stringfellow 1991), and no spectroscopic data are presently available. Brown dwarf members of α Per are expected to be somewhat more luminous than their Pleiades counterparts (about 30 Myr older) and have had less time to escape from the cluster center. In this *Letter* we report *R* and *I* photometry plus follow-up moderate-resolution spectroscopy of a BD candidate in α Per. The object, referred to as Ap 0325+4853, was detected in one of the seven fields that we have examined so far in this cluster. The position of this BD candidate on the H-R diagram is compared with that of Pleiades candidates in the framework of present theoretical evolutionary tracks and isochrones. We also propose a potentially powerful spectroscopic test for discriminating between very low mass stars (VLMs) and brown dwarfs.

2. OBSERVATIONS

The object presented in this *Letter* was found on two 2 × 2.9 arcmin^2 direct CCD images centered at the position of the star Ap 4 ($V = 16.1$, $V-I = 1.15$; Stauffer et al. 1985) located in the α Per cluster. The field is within the α Per nucleus, at a distance of about 16′ from the center (Artyukhina 1972). An automatic finding procedure detected Ap 0323+4853 as the 44th point source in decreasing order of brightness out of a total number of 61 detections in the *R*-filter CCD frame. In Figure 1 (Plate L9) we show this *R* image as a finding chart. The relatively bright star in the central region is Ap 4. The position of Ap 0323+4853, on the upper left corner (SW from Ap 4), is marked with an arrow. In Table 1 we give coordinates for Ap 0323+4853 derived from the offset measured on the CCD frames with respect to Ap 4.

The two CCD frames, one in *R* and one in *I*, where Ap 0323+4853 was discovered, are part of an ongoing survey program of deep CCD imaging in selected areas of young stellar associations. They were taken at the Cassegrain focus of the 1 m Jacobus Kapteyn Telescope at La Palma under dark photometric conditions. The survey sample, data reduction procedures, and photometric analysis will be described in Martín et al. (1992). Instrumental magnitudes for Ap 0323+4853 were derived by aperture photometry, and conversion to the standard *RI* Kron system was ensured through a calibration of three α Per stars, including Ap 4 itself, with known photometry (Stauffer et al. 1985) and seven red dwarfs from Hartwick, Cowley, & Mould (1984), observed on the same night (1991 January 16). Since most color to $T_{\rm eff}$ calibrations use $(R-I)_C$, we used the Bessell & Weis (1987) conversion from the Kron filters to the Cousins *RI* system. The results of the photometric analysis are given in Table 1.

The spectroscopic observations were conducted on 1991 September 16, using the 2.5 m Isaac Newton Telescope with the Intermediate Dispersion Spectrograph coupled to the 500

[1] Based on observations made with the Isaac Newton Telescope and the Jacobus Kapteyn Telescope, operated on the island of La Palma by the Royal Greenwich Observatory in the Spanish Observatorio del Roque de los Muchachos of the Instituto de Astrofisica de Canarias.

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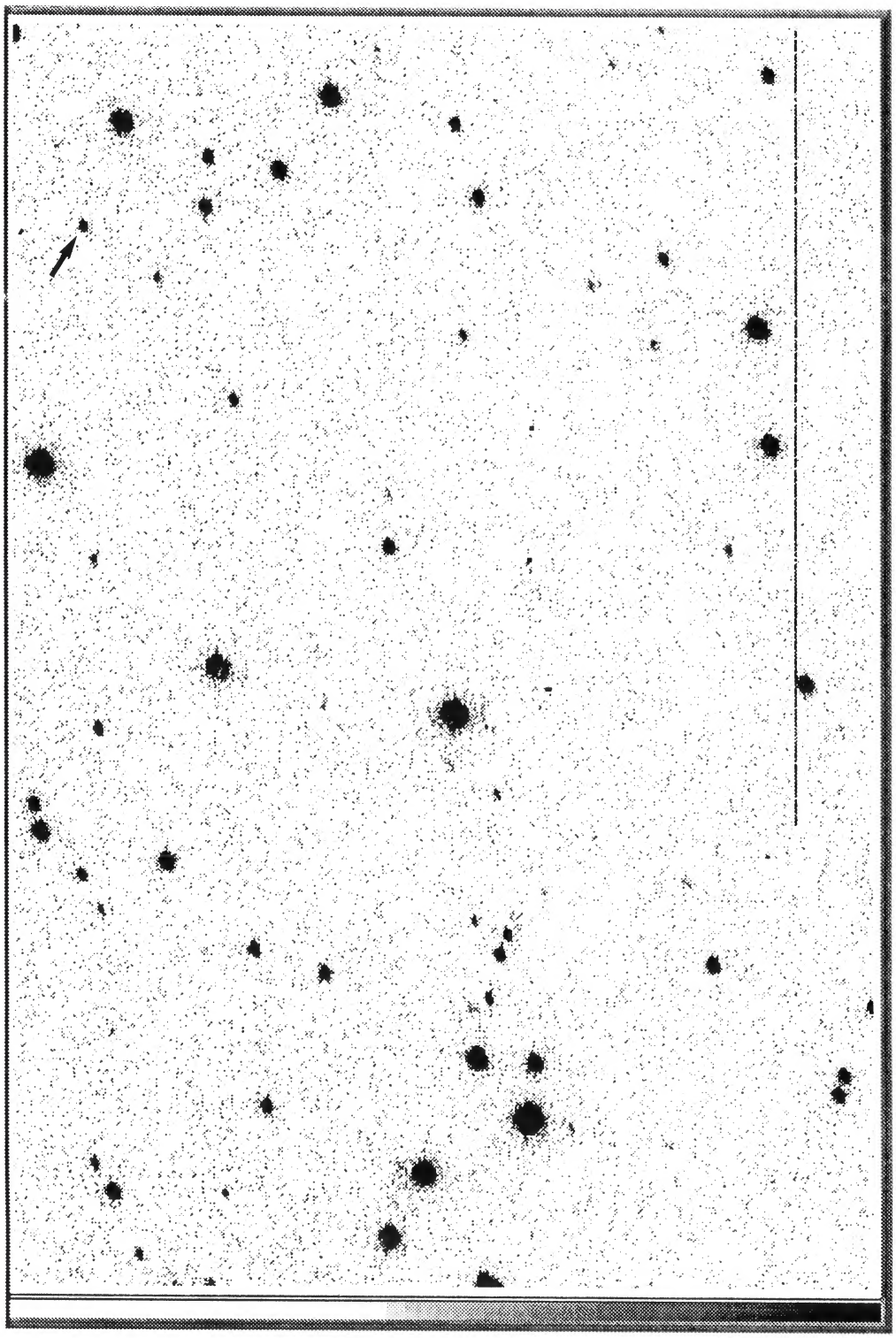


FIG. 1.—The 2 × 2.9 arcmin^2 *R*-filter CCD image where Ap 0323+4853 was discovered (arrow). The spatial resolution is 0.″3 pixel^{-1}. West is upward, and north is rightward.

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TABLE 1

Ap 0323+4853 DATA

Parameter	Value
R.A. (2000.0)	$03^h23^m03^s$
Decl. (2000.0)	48°53′06″
I_{Kron}	17.3 ± 0.1
$(R-I)_K$	1.73 ± 0.15
$(R-I)_C$	2.09 ± 0.15
J.D. obs.	2,448,515.684
$W_{H\alpha}$ (nm)	2.4 ± 0.2
J.D. obs.	2,448,515.703
$W_{H\alpha}$ (nm)	1.5 ± 0.3
J.D. obs.	2,448,515.752
$W_{H\alpha}$ (nm)	≤ 0.8
T_{eff} (K)	2840 ± 120
BC_I	0.01 ± 0.25
$\log (L/L_\odot)$	-2.6 ± 0.15

mm camera and a coated GEC CCD detector. The grating was the H150V, centered at 705 nm giving a spectral coverage of 163 nm and reciprocal dispersion of 0.28 nm pixel^{-1}. Effective resolution was about 0.9 nm. The excellent atmospheric conditions allowed us to identify all the objects in the field of Ap 0323+4853 (shown in Fig. 1) on the monitor of the guiding TV camera using the maximum on-line integration rate available. Thus, we are confident about the identification of the source. Two 1500 and one 1200 s exposures of Ap 0323+4853 were taken successively in order to avoid accumulation of cosmic-ray events. Each single spectogram was reduced via standard procedures (bias-correction, flat-fielding, and sky-subtraction) with the IRAF package[2] at the Instituto de Astrofísica de Canarias. The emission spectrum of a CuAr lamp, recorded before and after the object exposures, was used for wavelength calibration. The spectrum of the B5 flux standard star Hz 15 (Stone 1977) observed after the last object exposure at similar aimass was used for removal of telluric absorption bands and flux calibration.

[2] IRAF is distributed by National Optical Astronomy Observatory, which is operated by the Association of Universities for Research in Astronomy, Inc., under contract with the National Science Foundation.

3. PROPERTIES OF Ap 0323+4853

Our spectrum of Ap 0323+4853 is characterized by deep molecular band heads (TiO, VO, CaH), strong Hα in emission, and the presence of KI absorption, a feature usually taken as a gravity discriminant (Bessell 1991; Kirkpatrick, Henry, & McCarthy 1991) indicating that our object is a dwarf rather than a giant. In Figure 2 we present the resulting spectrum of Ap 0323+4853 after co-adding our three spectra. A two-point boxcar smoothing was applied in order to improve the signal-to-noise ratio and degrade our resolution for comparison with the spectrum of the extremely cool object PC 0025+047, kindly provided by D. P. Schneider. Schneider et al. (1991) show that the spectrum of PC 0025+047 is similar, perhaps even cooler than LHS 2924 and LHS 2930 (Probst & Liebert 1983) mainly because of deeper VO absorption and weaker TiO bands. Following the sequence proposed by the last authors and also Schneider et al. (1991), the deep TiO band heads and shallow VO of Ap 0323+4853 imply that this object is not as cool as PC 0025+047, but similar to LHS 2924. Prosser, Stauffer, & Kraft (1991) show that TiO bands strengths correlate well with spectral types for Pleiades and Hyades M dwarfs in the range M0–M5. The most sensitive ratio is their R_4/R_7, which decreases monotonically with later spectral type, reaching a value of 0.6 for M5.5. We obtain $R_4/R_7 = 0.64 \pm 0.06$ and 0.78 ± 0.07 for Ap 0323+4853 and PC 0025+047, respectively. Should the relationship of R_4/R_7 with spectral type continue to decrease for spectral types later than M5.5, it would mean that PC 0025+047 has earlier type than M4, entering in contradiction with its colors and the similarity with LHS 2924. Thus, the curve R_4/R_7 versus spectral type probably has an inflexion at about M6–7. An inflexion at about M7 also appears to be present in the color ratios data presented in Figures 6 and 7 of Kirkpatrick et al. (1991). The result of measuring their A and B bands on the spectrum

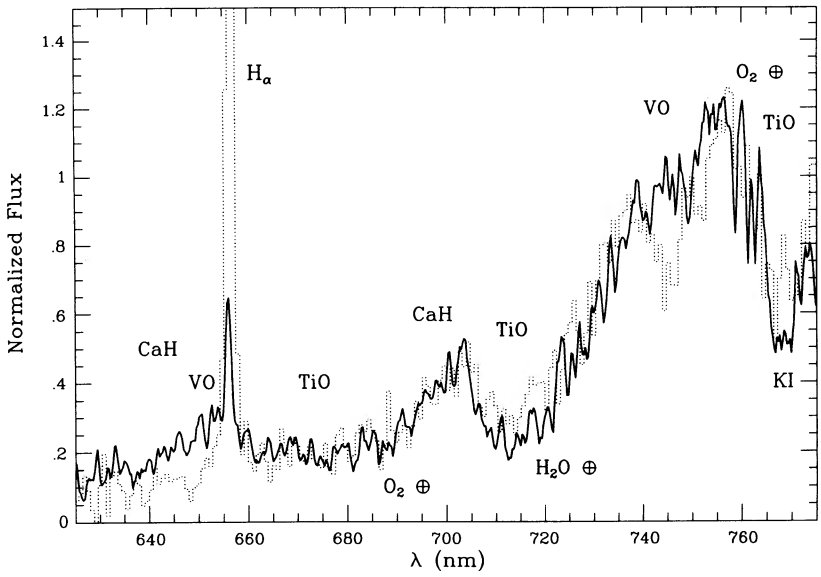


FIG. 2.—The weighted sum of our three spectrograms of Ap 0323+4853 (*solid line*) and the spectrogram of the BD candidate PC 0025+047 from Schneider et al. (1991) (*dotted line*). The main spectral lines and absorption band heads have been identified. The data are plotted in units of F_ν normalized to the pseudocontinuum at 750 nm. A two-point boxcar smoothing has been performed on our spectrogram in order to make our spectral resolution similar to that of Schneider et al. (1991).

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of Ap 0323+4853 (1.30 and 1.14, respectively) is consistent with our object having a spectral type M7–M8 (in Kirkpatrick et al.'s spectral classification system), and slightly lower gravity than the dwarfs used in their paper. To summarize, we estimate that the spectral type of Ap 0323+4853 is later than M5, probably M6–7, but limited spectral range and the paucity of reference stars with well-known gravity and metallicity impede a finer typification. Our $(R-I)_{\rm Kron}$ color is consistent with this type estimate as it is significantly redder than that of M5 stars in the Pleiades (Prosser et al. 1991).

A remarkable feature in the spectrum of Ap 0323+4853 is the prominent Hα emission and its rapid variability (see Fig. 3). In Table 1 we give the measured values of Hα equivalent widths. As indicated by the Julian Dates, the variations are significant in a time scale less than 1 hr. Unfortunately, the last spectrogram—where Hα is fading—is the noisiest of all three, and we can only place a 2 σ upper limit of ≈0.8 nm to the Hα strength. The behavior of Hα in Ap 0323+4853 could indicate flare activity, which in turn is an indication of youth. The presence of Hα in emission has been used as a criterion for low-mass membership to the Pleiades cluster (Prosser et al. 1991). Since the α Per cluster is slightly younger than the Pleiades, we feel justified in applying the same criterion to Ap 0323+4853. The value of W_λ(Hα) in our first spectrogram (see Table 1) is a factor of 2 above the highest values measured in the Pleiades, and it is similar to the Hα of ESO 207-61, the brown dwarf candidate associated with the Hyades moving group (Ruiz et al. 1991). The upper limit of Hα equivalent width for our third spectrogram is similar to the maximum values of Hα strengths for Pleiades members of spectral types M4–5 (Prosser et al. 1991). The Hα centroid velocities are consistent within the measurement error bars (±70 km s^{-1}) with the radial velocities of known α Per members.

On the basis of the previous analysis, we assume that our object is a cluster member for location on the H-R diagram. To derive $T_{\rm eff}$ we used the $(R-I)_{\rm C}$ versus $(V-I)_{\rm C}$ relationship of Bessell (1991) [assuming $E(V-I)_{\rm K}=0.16$ as the average reddening toward α Per; Stauffer et al. 1985], and the calibration $(V-I)_{\rm C}$ versus $T_{\rm eff}$ given in Stringfellow (1991). In Table 1, we give the effective temperature, bolometric correction, and luminosity obtained using the calibrations by the last author. The ±120 K error bar in effective temperature given in the table is mainly a consequence of the photometric uncertainties. Since Bessell (1991) notes that $R-I$ may not be sensitive to temperature for spectral types later than about M6, and because a spectral type as late as M7 could be appropriate for Ap 0323+4853, we consider adequate to enlarge the cool part of this error bar to include the $T_{\rm eff}$ corresponding to such a spectral type (~2600 K). The uncertainty in luminosity comes from the combination of uncertainties in the photometry, bolometric correction, and α Per distance. In Figure 4 we have plotted Ap 0323+4853 together with possible Pleiades BDs from Stauffer et al. (1989); the α Per BD candidate shares the same region of the H-R diagram as the Pleiades candidates.

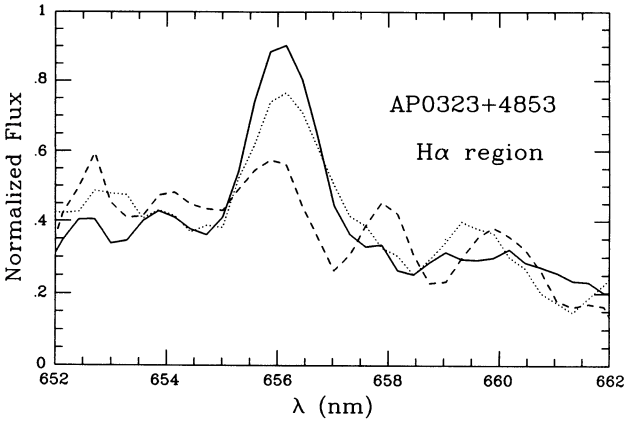


FIG. 3.—Variations of the Hα emission in Ap 0323+4853. The Julian dates of the observations are given in Table 1. The time interval between our first spectrum (*solid line*) and the second (*dotted line*) was ≈½ hr, and between the second and the third (*dashed line*) was ≈1 hr.

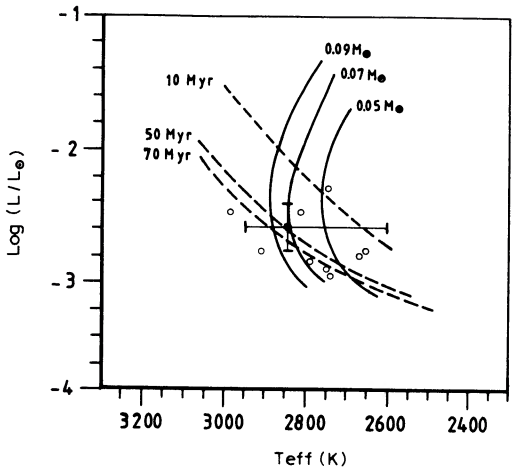


FIG. 4.—Our BD candidate in the H-R diagram (*filled circle*). The error bars are explained in the text. We also plot the possible BDs in the Pleiades (*open circles*) according to Stringfellow (1991) for comparison with our object. Superposed to the data are evolutionary tracks (*solid lines*) and isochrones (*dotted lines*) from the last author.

Superposed on the data in Figure 4 we have drawn theoretical evolutionary tracks for masses between 0.09 and 0.05 $M_\odot$ and isochrones for 10, 50, and 70 Myr from Stringfellow (1991). The mass and age of Ap 0323+4853 in the grid of Stringfellow are ~0.07 $M_\odot$ and 50 Myr. However, given the obervational error bars, its mass could be in the range 0.12–0.025 $M_\odot$. We also note that using the evolutionary models of D'Antona & Mazzitelli (1985) or Nelson, Rappaport, & Joss (1986), we obtain a lower mass by a factor ~2 and smaller age by a factor 2–3. Bearing in mind the difficulties modeling young brown dwarfs (see, e.g., Stevenson 1991) and the errors involved in the determination of oberved parameters for our BD candidate, at present we cannot definitely establish whether or not our object is substellar. Moreover, although all the observational evidence accumulated so far (location, strong Hα emission, rapid variability) supports membership of Ap 0323+4853 in the α Per cluster, we do not have a definitive proof and cannot completely rule out that this object might be a foreground flare M6–7 dwarf that happens to lie along the line of sight of the cluster at the right distance from us to have the apparent magnitude that would correspond to a young contracting BD in α Per. In order to discriminate between the two possibilities, i.e., a BD member of α Per or a VLM star in the field, in the following section we discuss a possible spectroscopic test.

4. THE LITHIUM TEST

The lithium surface abundance is widely used as a probe of conditions inside the stars. This light element burns at temperatures of about 2.5 × 10^6 K, easily attained even in the interiors of very low mass M stars. In fact, as far as we know,

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no detection of lithium in main-sequence M stars has been reported, and only in very young T Tauri stars of this spectral type the Li I doublet has been observed. Observations of M-type members of the Pleiades (García López et al. 1990) and α Per (García López et al. 1992) show that they have depleted lithium by a factor $\geq 10^3$. Such a short time scale for lithium destruction is in agreement with theoretical predictions (Pinsonneault, Kawaler, & Demarque 1990) and explains the nondetections of lithium in main-sequence M stars.

As noted by Rebolo (1991) and Magazzù, Martín, & Rebolo (1991), the situation for objects with masses below the substellar limit can be very different. Calculations by several groups (D'Antona & Mazzitelli 1985; Nelson et al. 1986; Burrows, Hubbard, & Lunine 1989) show that for masses under $\sim 0.06\ M_\odot$ the maximum central temperature is always below the lithium-burning temperature. Consequently, lithium should be preserved in substellar objects. Pozio (1991) has modeled tentatively the evolution of lithium in brown dwarfs and finds that no significant lithium depletion occurs below $0.06\ M_\odot$, even though the high densities in their interiors would allow the destruction of lithium at slightly lower temperatures (1.8×10^6 K) than the typical value mentioned above. Given the opposite behavior of lithium in VLM stars and BDs, and considering the surface temperature and pressure of objects near the substellar mass limit, we claim that observations of the Li I resonance doublet at 670.8 nm can be used successfully as a powerful spectroscopic test for discriminating between these two types of objects.

Our claim is supported by computations of the Li I doublet equivalent widths for BD candidates using model atmospheres from the grid of Allard (1990). For $T_{\rm eff}$ in the range 2750–2000 K, $\log g = 4.0$–5.0, solar metallicity, and a cosmic lithium abundance [$\log N(\mathrm{Li}) = 3.10$], we find strong lithium lines, $W_{\rm Li}$, of 0.15–0.25 nm. The LTE calculations are quite sensitive to the adopted opacities, so we cannot discard corrections in our equivalent widths by a factor 2. Preliminary calculations indicate that the NLTE corrections are not larger than 10%–20% (Ya. V. Pavlenko, private communication). Therefore if lithium is preserved in BDs, such intense lines make its detection possible. Unfortunately, for the candidate presented in this letter, the effective resolution (~ 0.8 nm) and S/N ratio (less than 10 in the lithium region) of our spectrum sets the 1 σ detection limit at $W_\lambda \sim 0.24$ nm, hindering consideration on its lithium abundance. With higher resolution (~ 0.1 nm) and slightly higher S/N, the lithium test can be applied to our BD candidate and other candidates in the Pleiades and in the field. We note that observations reported by Magazzù et al. (1991) of lithium in the very low mass pre–main-sequence object UX Tau C ($T_{\rm eff} \approx 3000$ K) show the feasibility of the test proposed here.

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